[Blinkit Data Analysis]

This project aims to bridge the gap between raw operational data and actionable business insights using analytics and visualization.

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Blinkit Data Analysis Project Report

1. Business Statement

Blinkit operates in a highly competitive quick-commerce (q-commerce) ecosystem where speed, availability, customer experience, and cost efficiency directly impact customer retention and profitability. With thousands of daily orders, multiple product categories, time-bound deliveries, and continuous marketing campaigns, Blinkit generates large volumes of operational and transactional data. However, raw data alone does not provide actionable insights unless it is structured, cleaned, and analyzed systematically.

With growing customer expectations and increasing order volumes, Blinkit requires a **centralized analytics solution** to monitor performance across sales, customers, inventory, delivery operations, and marketing campaigns.

The business needs a **data-driven approach** to:

- Understand customer purchasing behavior
- Identify high-value and at-risk customers
- Optimize inventory to prevent stock-outs and losses
- Evaluate delivery performance and its impact on customer satisfaction
- Measure marketing effectiveness and ROI

This project aims to bridge the gap between **raw operational data and actionable business insights** using analytics and visualization.

2. Problem Statement

Blinkit data is spread across multiple operational systems, including customer records, orders, products, inventory, delivery performance, marketing campaigns, and customer feedback.
However, the business faces challenges such as:

- Lack of a **single consolidated view** of customers, sales, and products
- Difficulty in identifying **top revenue drivers and profit leakages**
- Limited visibility into **inventory health and damaged stock**
- Inability to easily assess how **delivery delays impact customer ratings**
- Challenges in tracking **customer satisfaction and sentiment trends**
- Inefficient evaluation of **marketing campaign performance and ROAS**
- Absence of an interactive dashboard for **real-time decision-making**

Because of these issues, stakeholders struggle to take **timely, informed, and strategic decisions**.

3. Objective of the Analysis

The objective of this project is to:

- Build an **end-to-end analytics pipeline** using
  Python (EDA & ETL) → SQL (Business Queries) → Power BI (Visualization)
- Create a **master analytical dataset** for flexible analysis
- Perform **EDA** to understand data distribution, trends, and relationships
- Use **SQL** to answer real-world business questions related to:
  - Revenue
  - Customers
  - Products
  - Inventory
  - Delivery performance
  - Customer satisfaction
- Develop an **interactive Power BI dashboard** that enables:
  - Performance monitoring
  - Drill-down analysis
  - Strategic and operational decision-making

4. Data Overview

The analysis uses multiple structured datasets representing different business functions. Below is an overview of all datasets, their purpose, and key columns.

4.1 Customers Dataset (blinkit_customers)

Records: 25,000

Key Columns

- customer_id
- customer_name
- email, phone, address
- area, pincode
- registration_date
- customer_segment
- total_orders
- avg_order_value

Purpose: Used to analyze customer demographics, segmentation, and overall customer value.

4.2 Orders Dataset (blinkit_orders)

Records: 50,000

Key Columns:

- order_id
- customer_id
- order_date
- promised_delivery_time
- actual_delivery_time
- delivery_status
- order_total
- payment_method
- delivery_partner_id
- store_id

Purpose: Captures transactional order-level data for revenue, order trends, and delivery fulfillment analysis.

4.3 Order Items Dataset (blinkit_order_items)

Records: 225,238

Key Columns:

- order_id
- product_id
- quantity
- unit_price
- total_price

Purpose: Enables granular product-level sales analysis and basket composition insights.

4.4 Products Dataset (blinkit_products)

Records: 237

Key Columns:

- product_id
- product_name
- category
- brand
- price
- mrp
- margin_percentage
- shelf_life_days

- min_stock_level
- max_stock_level

Purpose: Provides product master data for category, pricing, and margin-based analysis.

4.5 Inventory Dataset (blinkit_inventory)

Records: 259,752

Key Columns:

- product_id
- date
- stock_received
- damaged_stock

Purpose: Tracks daily inventory movement and damage for stock health and wastage analysis.

4.6 Delivery Performance Dataset (blinkit_delivery)

Records: 50,000

Key Columns:

- order_id
- delivery_partner_id
- promised_time
- actual_time
- delivery_time_minutes
- distance_km
- delivery_status
- reasons_if_delayed

Purpose: Used to analyze delivery efficiency, delays, and operational bottlenecks.

4.7 Marketing Performance Dataset (blinkit_marketing)

Records: 9,864

Key Columns:

- campaign_id
- campaign_name
- date
- target_audience
- channel
- impressions

- clicks
- conversions
- spend
- revenue_generated
- roas

Purpose: Evaluates marketing campaign effectiveness and return on ad spend.

4.8 Supporting Datasets

- **Category Icons (blinkit_category_icon)**: Category-to-icon mapping
- **Rating Icons (blinkit_rating_icon)**: Rating visualization support

These datasets are primarily used for dashboard visualization enhancement.

5. Data Merging & Feature Engineering

5.1 Inventory Aggregation

To create a consolidated inventory view, the raw inventory data was aggregated at a **product-date level**:

- stock_received aggregated using sum
- damaged_stock aggregated using sum

This ensured accurate daily inventory movement without duplication.

5.2 Inventory Master Table Creation

The aggregated inventory data was merged with selected product attributes:

- product_name
- category
- brand
- price

This resulted in a comprehensive **Inventory Master Table** with:

- Shape: **259,752 rows × 8 columns**
- No duplicate records at product_id + date level

This table forms the foundation for inventory dashboards and stock analysis.

5.3 Blinkit Master Table Creation (df_blinkit)

1. Objective of Creating the Blinkit Master Table

The purpose of creating the **Blinkit Master Table (df_blinkit)** was to consolidate all essential business data into **one single analytical table**.
This approach avoids repeated joins across multiple files and allows **easy, flexible analysis** for EDA, SQL queries, and Power BI dashboards.

2. Data Aggregation & Merging Process

To create the master table, multiple datasets were merged step by step using **left joins**, ensuring no loss of transactional data:

- **Customers + Orders** → customer and order-level information
- **Order Items** → product quantity and unit price
- **Products** → category, brand, pricing, and margin details
- **Customer Feedback** → ratings, sentiment, and feedback text

After merging, aggregation was performed at the **Customer–Order level** to:

- Remove duplication caused by multiple products per order
- Summarize quantities correctly
- Calculate average ratings per order
- Retain only business-relevant attributes

Additional enrichment was done by merging:

- **Category icons** (for dashboard visuals)
- **Rating emojis and stars** (for better rating interpretation)

3. Final Columns in Blinkit Master Table

The final master table contains the following columns:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 35 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   customer_id           50000 non-null  int64
 1   order_id              50000 non-null  float64
 2   customer_name         50000 non-null  object
 3   email                 50000 non-null  object
 4   phone                 50000 non-null  int64
 5   address               50000 non-null  object
 6   pincode               50000 non-null  int64
 7   customer_segment      50000 non-null  object
 8   total_orders          50000 non-null  int64
 9   avg_order_value       50000 non-null  float64
 10  actual_delivery_time  50000 non-null  object
 11  delivery_status       50000 non-null  object
 12  order_total           50000 non-null  float64
 13  payment_method        50000 non-null  object
 14  delivery_partner_id   50000 non-null  float64
 15  product_id            50000 non-null  float64
 16  quantity              50000 non-null  float64
 17  unit_price            50000 non-null  float64
 18  product_name          50000 non-null  object
 19  category              50000 non-null  object
 20  brand                 50000 non-null  object
 21  price                 50000 non-null  float64
 22  mrp                   50000 non-null  float64
 23  margin_percentage     50000 non-null  float64
 24  min_stock_level       50000 non-null  float64
 25  max_stock_level       50000 non-null  float64
 26  feedback_id           50000 non-null  float64
 27  rating                50000 non-null  float64
 28  feedback_text         50000 non-null  object
 29  feedback_category     50000 non-null  object
 30  sentiment             50000 non-null  object
 31  last_orderdate        50000 non-null  object
 32  category_icon         50000 non-null  object
 33  rating_emoji          50000 non-null  object
 34  rating_star           50000 non-null  object
```

Customer & Profile Information

- customer_id
- customer_name
- email
- phone
- address
- pincode
- customer_segment
- total_orders
- avg_order_value

Order & Delivery Information

- order_id
- last_orderdate
- order_total
- payment_method

- delivery_status
- actual_delivery_time
- delivery_partner_id

Product & Pricing Information

- product_id
- product_name
- category
- brand
- price
- mrp
- margin_percentage
- quantity
- unit_price
- min_stock_level
- max_stock_level
- category_icon

Customer Feedback & Ratings

- feedback_id
- rating
- rating_star
- rating_emoji
- feedback_text
- feedback_category
- sentiment

4. Output After Master Table Creation

- **Granularity:** One row per **Customer per Order**
- **Total Columns:** 35
- **Duplicate Records:** None (validated after aggregation)
- **Missing Values:** No critical missing values impacting analysis
- **Data Quality:** Clean and analysis-ready

The dataset was validated to ensure:

- No duplicate customer–order records

- Consistent column naming
- Correct aggregation logic

5. Why This Master Table Was Created

The **Blinkit Master Table** acts as the **core analytical foundation** of the project.
It enables:

- Efficient **EDA** without repeated joins
- Faster and cleaner **SQL business queries**
- Smooth integration with **Power BI dashboards**
- Consistent logic across all analysis layers

By centralizing customer, order, product, and feedback data into one table, the analysis becomes **scalable, reliable, and business-focused**.

6. Data Cleaning & Quality Checks

Comprehensive data quality checks were performed across all datasets:

- **Missing Values:** No critical missing values found
- **Duplicates:** No duplicate records detected after aggregation and merges
- **Outliers:** Minor outliers identified but deemed business-valid and not impactful
- **Column Standardization:** Column names sanitized to ensure SQL compatibility

As a result, the data is considered **clean, reliable, and analysis-ready**.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand data distributions, customer behavior, product performance, and relationships between key variables. The analysis includes **univariate** and **bivariate** visualizations, followed by correlation analysis. The insights below are based on the graphs generated during the analysis phase.

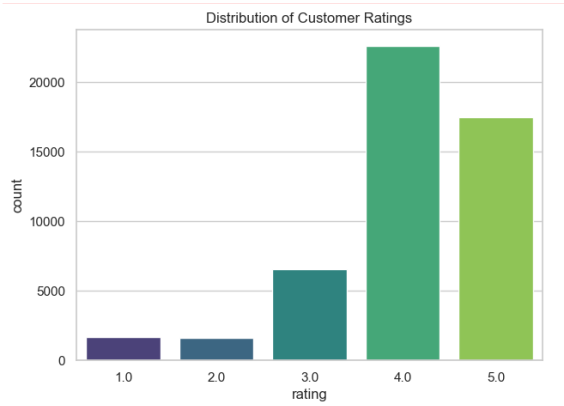
7.1 Univariate Analysis

7.1.1 Distribution of Customer Ratings

What was analyzed:
The distribution of customer ratings to understand overall customer satisfaction.

Key observations:

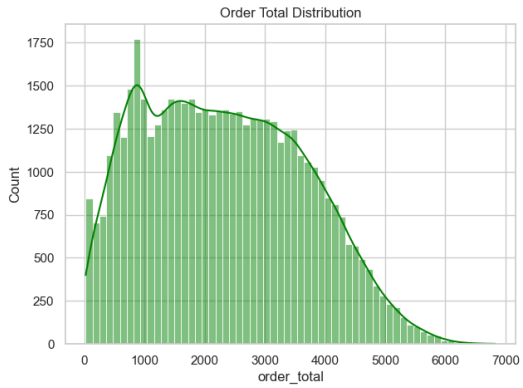
- Rating **4** is the most frequent rating in the dataset
- Rating **5** is the second most common
- Rating **3** follows next



Insight:
Most customers are satisfied with Blinkit's service, as ratings are skewed towards the higher end. Extremely low ratings are relatively limited, indicating generally positive customer experience.

7.1.2 Order Total Distribution

What was analyzed:
Distribution of order values to identify typical customer spending behavior.



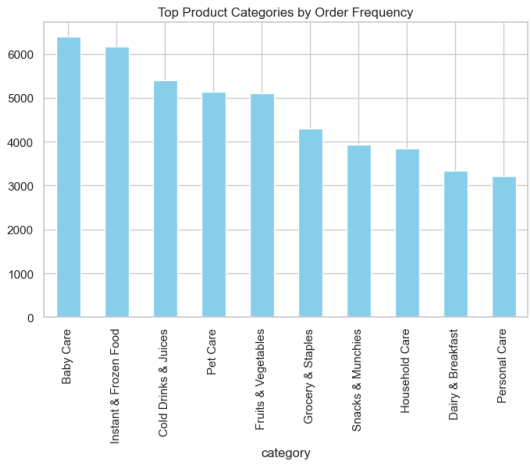
Key observations:

- The highest concentration of orders lies between **₹1250 and ₹1500**
- Very high-value orders exist but occur less frequently

Insight:
The majority of customers place mid-range orders, which can be considered the standard basket size for Blinkit. This range can be used as a reference for pricing strategies and promotional offers.

7.1.3 Top Product Categories

What was analyzed:
Category-wise frequency to identify the most popular product categories.



Top categories identified:

1. Baby Care
2. Instant & Frozen Food
3. Cold Drinks & Juices

Insight:
These categories form the backbone of Blinkit's demand and should be prioritized for inventory planning, promotions, and supplier optimization.

7.1.4 Customer Segment vs Rating

What was analyzed:
Distribution of ratings across customer segments (Inactive, New, Regular, Premium).

Observation:

- No significant variation in ratings across customer segments

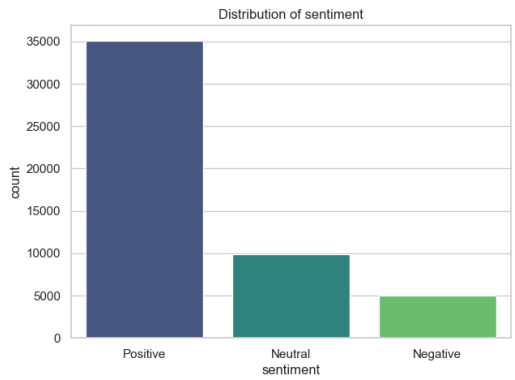
Insight:
Customer satisfaction levels remain relatively consistent irrespective of customer segment. This indicates uniform service quality across different types of customers.

7.1.5 Sentiment Distribution

..**What was analyzed:**
Sentiment classification of customer feedback.

Key observations:

- Positive sentiments: ~35,000 records
- Neutral sentiments: Moderate volume
- Negative sentiments: ~5,000 records



Insight:
The dominance of positive sentiment reflects strong customer satisfaction. The smaller proportion of negative feedback provides targeted opportunities for service improvement.

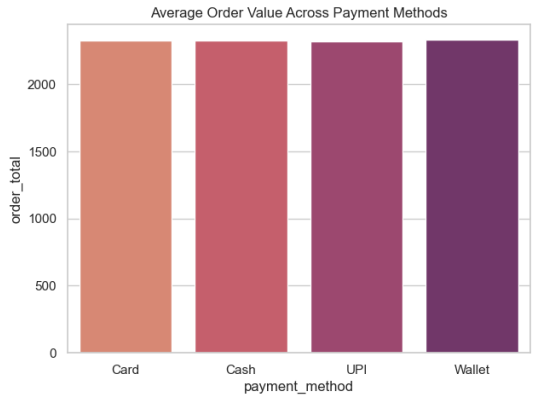
7.2 Bivariate Analysis

7.2.1 Payment Method vs Order Total

What was analyzed:
Comparison between payment methods and order values.

Observation:

- No significant difference in order total across different payment methods



Insight:
Customer spending behavior is not influenced by payment mode, indicating flexibility and consistent purchasing patterns across all payment options.

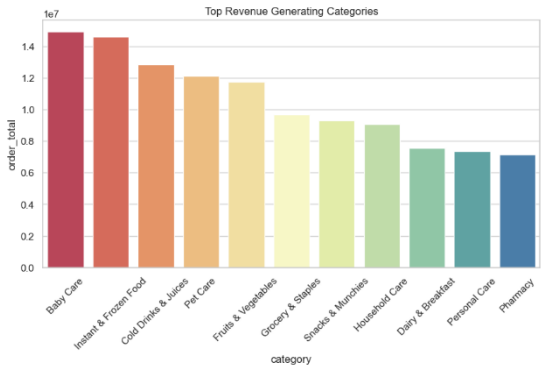
7.2.2 Product Category vs Order Total

What was analyzed:
Revenue contribution by product category.

Key findings:

- Baby Care
- Instant & Frozen Food
- Cold Drinks & Juices

These categories generate the highest total order value.



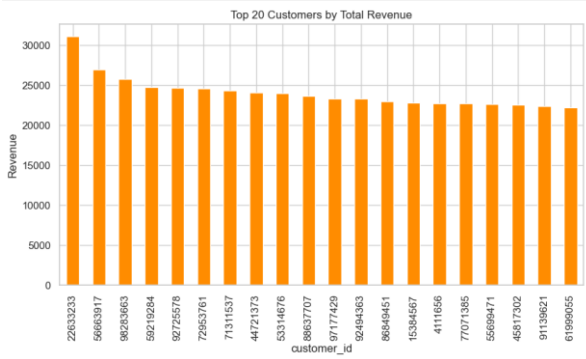
Insight:
These categories are the primary revenue drivers and should be central to category-level growth strategies.

7.2.3 Top 20 Customers by Total Revenue

What was analyzed:
Customer-wise revenue ranking using customer IDs.

Observation:

- A small group of customers contributes a disproportionately high share of revenue



Insight:
High-value customers represent a strong opportunity for loyalty programs, personalized offers, and retention strategies.

7.2.4 Margin Percentage vs Price

What was analyzed:
Relationship between product price and margin percentage.

Insight:
Understanding this relationship helps in optimizing pricing strategies without negatively impacting profitability.

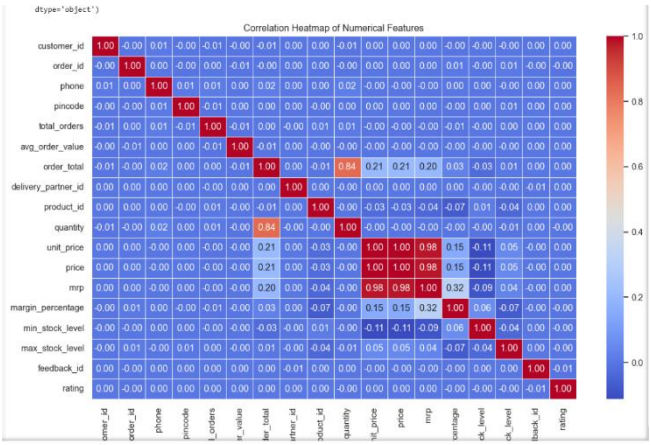
7.3 Correlation Analysis

7.3.1 Correlation Heatmap

What was analyzed:
Correlation among numerical variables such as price, MRP, quantity, and margin.

Key correlations identified:

- **MRP and Price** show a strong positive correlation
- **Unit Price and MRP** are highly correlated
- **Quantity and Price** also show strong correlation



Insight:
These strong correlations confirm logical pricing structures and consistent product pricing behavior. Pricing and quantity relationships can be leveraged for demand forecasting and promotional modeling.

8. EDA Summary

- Customer satisfaction is generally high
- Order values are concentrated in a mid-range band
- A few product categories drive most of the revenue
- High-value customers contribute significantly to overall sales
- Pricing and MRP are structurally aligned

These insights provide a strong foundation for **SQL-based problem solving and dashboard-driven decision making**.

9. SQL Database Integration

All cleaned datasets were successfully migrated from Jupyter Notebook to a MySQL database (df_blinkit) using SQLAlchemy and PyMySQL. Each dataset was stored as a separate table, including the final inventory_master table.

Objective of SQL Integration:

- Enable complex analytical queries
- Support problem-statement-driven analysis
- Prepare structured data for dashboard and reporting layers

10. SQL Analysis: Business Questions & Explanations

This section documents the **business-driven SQL analysis** performed on the df_blinkit_table.
Each query is designed to answer a specific operational or strategic question using structured SQL logic.

Q. 1 Top 10 Customers by Total Revenue

Business Question:
Who are the top 10 customers based on total revenue generated?

Purpose of the analysis:
To recognize high-value customers who significantly impact overall revenue.

Why this matters:
These customers are ideal candidates for:

- Loyalty programs
- Personalized offers
- Premium membership benefits
- Retention-focused strategies

Result Grid | Filter Rows:

customer_id	total_product_purches
22633233	31058.480000000007
56663917	26933.190000000002
98283663	25754.870000000003
59219284	24698.219999999998
92725578	24649.73
72953761	24536.06
71311537	24287.260000000002
44721373	24045.73
53314676	23996.1
88637707	23662.79

Q. 2 Product Categories Generating the Highest Revenue

Business Question:
Which product categories generate the highest total revenue?

Why this matters:
Helps in:

- Category-level investment decisions
- Inventory prioritization
- Marketing focus
- Supplier and pricing strategy optimization

category	total_revenue
Baby Care	14923998.019999992
Instant & Frozen Food	14618860.240000015
Cold Drinks & Juices	12833275.169999959

Q. 3 Most Used Payment Method

Business Question:
Which payment method is used most frequently by customers?

Purpose of the analysis:
To identify customer payment preferences.

Why this matters:
Supports:

total_payment	payment_method
12674	Wallet
12525	UPI
12416	Card
12385	Cash

- Payment partner optimization
- Cashback and discount planning
- Improving checkout experience

Q. 4 Products with the Highest Margin Percentage

Business Question:
Which products have the highest profit margins?

margin_percentage	product_name	product_id
40	et ex	616389
40	quibusdam quas	4104
40	voluptates provident	744975
40	exercitationem provident	854898
40	placeat consequatur	206296

Purpose of the analysis:
To identify highly profitable products.

Why this matters:
These products can be:

- Promoted aggressively
- Prioritized in recommendations
- Used to improve overall profitability

Q. 5 Most Ordered Products (Quantity-wise)

Business Question:
Which products are ordered the most based on quantity sold?

delivery_status	avg_rating
Slightly Delayed	3.5025
On Time	4.4955
Significantly Delayed	1.9674

Purpose of the analysis:
To understand product demand intensity.

Why this matters:
Helps with:

- Demand forecasting
- Stock replenishment planning
- Supply chain optimization

Q. 6 Customer Segment with the Highest Number of Orders

Business Question:
Which customer segment places the highest number of orders?

To understand engagement levels across customer segments.

Why this matters:
Supports:

- Segment-specific marketing strategies
- Customer lifecycle management
- Retention planning

Q. 7 Delivery Status vs Average Customer Rating

Business Question:
How does delivery performance affect customer ratings?

Purpose of the analysis:
To assess the impact of delayed deliveries on customer satisfaction.

customer_segment	total_orders
Inactive	12708
Regular	12549
Premium	12159
New	12584

Why this matters:
Highlights:

- Operational weaknesses
- Importance of timely delivery
- Areas for logistics improvement

Q. 8 Products with Low Stock but High Sales

Business Question:
Which products sell well but have low minimum stock levels?

Purpose of the analysis:
To detect products at risk of stock-outs.

Why this matters:
Helps prevent:

- Lost sales
- Customer dissatisfaction
- Supply chain disruptions

Q. 9 High-Value Customers with Consistently Low Ratings (Churn Risk)

Business Question:
Which high-spending customers are dissatisfied?

Purpose of the analysis:
To detect potential churn risk among valuable customers.

customer_id	customer_name	total_spent	avg_rating	total_orders
13256345	Libni Amble	20685.51	2.8333333333333335	6
58453778	Unni Pandit	17846.5	2.8333333333333335	6
88927567	Gautami Rout	15489.57	2.75	4
68032743	Oviya Roy	13719.720000000001	2.5	4
23535376	Frederick Taneja	13565.91	2.8	5
28959971	Chavvi Dhar	13312.439999999999	2.8	5
11811405	Nidra Rattan	12907.73	2.75	4

Why this matters:
Enables:

- Proactive customer retention
- Personalized issue resolution
- Revenue protection strategies

Q. 10 Products with High Sales but Low Margins (Profit Leakage)

Business Question:
Which products sell well but generate low profit margins?

total_sold	average_margin_percentage	product_id	product_name
2145	15	607426	eos voluptatum
2106	15	901230	eos fuga
2082	15	687682	dolores quia
2056	15	469022	reprehenderit molestias
2047	15	270538	placeat eveniet
2020	15	446599	numquam corporis
2006	15	715324	quasi sapiente

Purpose of the analysis:
To identify profitability gaps.

Why this matters:
Supports:

- Pricing optimization

- Cost control
- Supplier renegotiation

Q. 11 Delivery Partners Whose Delayed Orders Receive Poor Ratings

Business Question:
Which delivery partners receive poor ratings for delayed deliveries?

Purpose of the analysis:
To assess operational accountability at the delivery partner level.

Why this matters:
Helps in:

delivery_partner_id	delayed_orders	avg_rating	delay_count
45608	1	1	1
45074	1	1	1
61028	1	1	1
11062	1	1	1
88607	1	1	1
33737	1	1	1
36935	1	1	1

- Performance benchmarking
- Partner evaluation
- Improving last-mile delivery quality

Q. 12 Product Categories Impacting Customer Satisfaction

category	total_orders	avg_rating	avg_spend	total_units_sold
Dairy & Breakfast	3332	4.020108043217287	2262.590288115247	29884
Personal Care	3211	4.02616007474307	2285.5507972594287	28448
Snacks & Munchies	3935	4.036848792884371	2364.827977128332	35446
Household Care	3854	4.044369486248054	2346.7344239750873	34651
Baby Care	6398	4.046108158799625	2332.603629259142	57935
Fruits & Vegetables	5110	4.047553816046967	2298.197825831694	46074
Pet Care	5131	4.056908984603391	2357.6996374975665	46116

Business Question:
Which product categories have the highest and lowest customer satisfaction?

What this query analyzes:
Compares average ratings and order volume across categories.

Why this matters:
Supports:

- Quality control initiatives
- Category-level experience improvements
- Strategic category planning

Blinkit Analysis – Power BI Dashboard Report

11. Dashboard Objective

The primary objective of the Power BI dashboard is to **convert cleaned and SQL-validated data into interactive visual insights** that help stakeholders understand:

- Business performance
- Customer behavior

- Product demand
- Inventory health
- Delivery efficiency
- Marketing effectiveness

The dashboard is designed for **decision-making**, not just visualization.

11.1 Data Source & Integration

- Data Source: **MySQL Server**
- Tables used:
    - df_blinkit_table (master transactional table)
    - inventory_master
    - marketing_performance
- Data refresh enabled via SQL Server connection
- Minimal transformations applied in Power Query since data was already cleaned in Jupyter & SQL

11.2 Dashboard Design & Navigation

- Custom Blinkit-themed UI (figma)
- Button-based navigation for smooth page switching
- Clean layout with:
    - KPI cards
    - Icons
    - Consistent color coding
- Filters and slicers placed logically for easy interaction

The design ensures **usability, clarity, and executive readability**.

12. Dashboard Pages & Key Insights

12.1 Overview Page

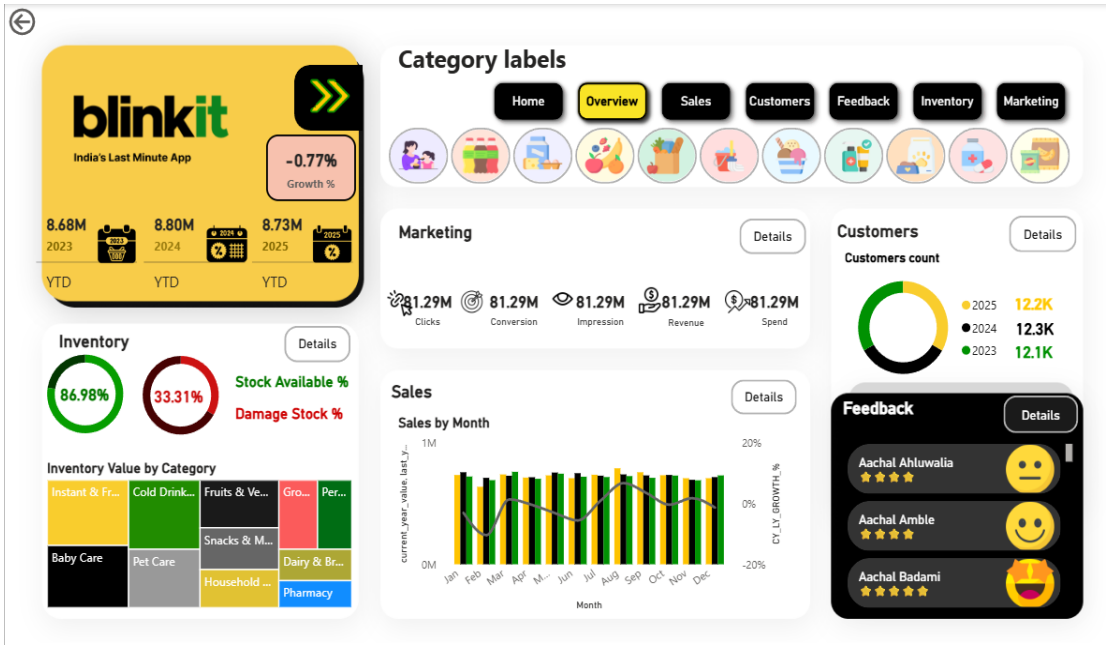
Purpose:
Provide a high-level snapshot of overall business performance.

Key Insights:

- Year-wise sales trend comparison
- Overall growth indicators
- Customer rating distribution using icons and emojis
- Inventory stock vs damaged stock summary

- Marketing performance overview (Impressions, Clicks, Conversions, ROAS)

This page acts as a **one-glance executive summary**.



12.2 Sales Page

Purpose:
Analyze revenue patterns and sales drivers.

Key Insights:

- Monthly and yearly sales trends
- Top revenue-generating product categories:
    - Baby Care
    - Instant & Frozen Food
    - Cold Drinks & Juices
- Payment method analysis showed no extreme dominance
- Top customers by revenue identified

- Geographic sales distribution using pincode-level data

Helps identify **what sells the most, when, and to whom**.

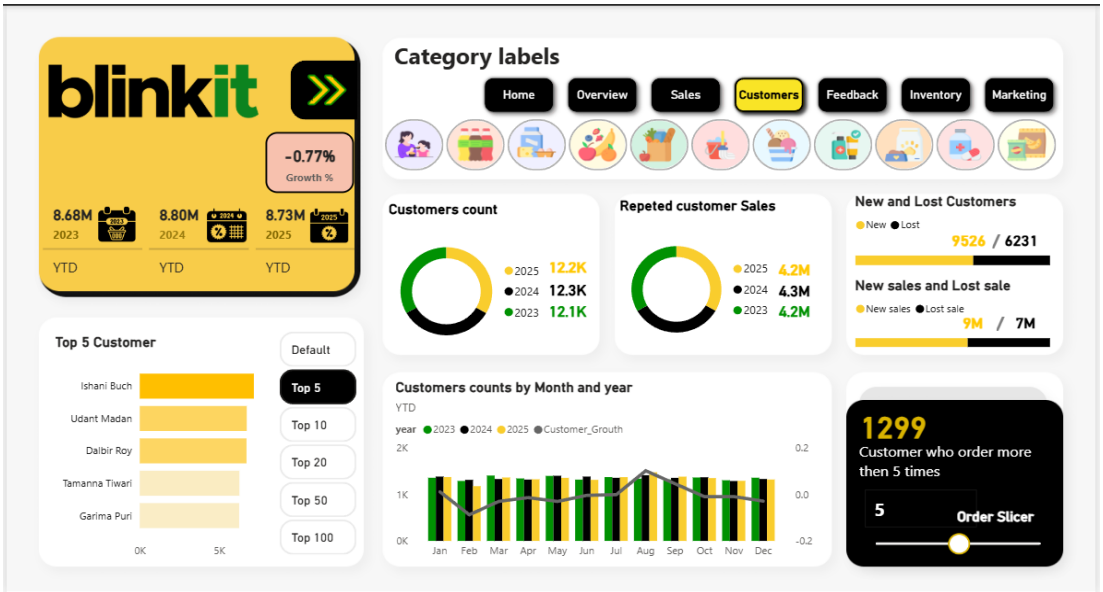
12.3 Customer Page

Purpose:
Understand customer growth, retention, and churn behavior.

Key Insights:

- Year-wise customer growth comparison
- New vs Lost customer analysis:
    - New customers significantly contribute to revenue
    - Lost customers indicate churn risk
- Repeat customers (5+ orders) identified
- Monthly customer acquisition trends

Supports **customer retention and lifecycle strategies**



12.4 Feedback & Rating Page

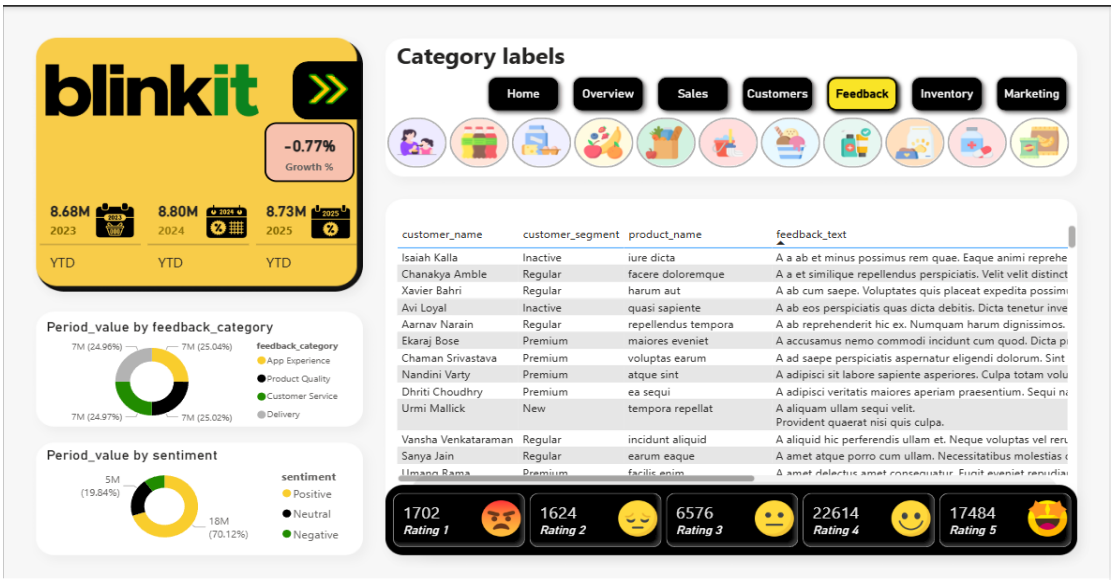
Purpose:
Analyze customer satisfaction and sentiment.

Key Insights:

- Rating distribution:
    - 4-star ratings dominate
    - Followed by 5-star and 3-star ratings
- Sentiment distribution:
    - Positive feedback ≈ 35,000

    - Neutral feedback moderate
    - Negative feedback lowest (~5,000)
- Feedback category analysis with customer comments
- Emoji and star-based rating visuals improve interpretability

Helps link **customer experience with business performance**.



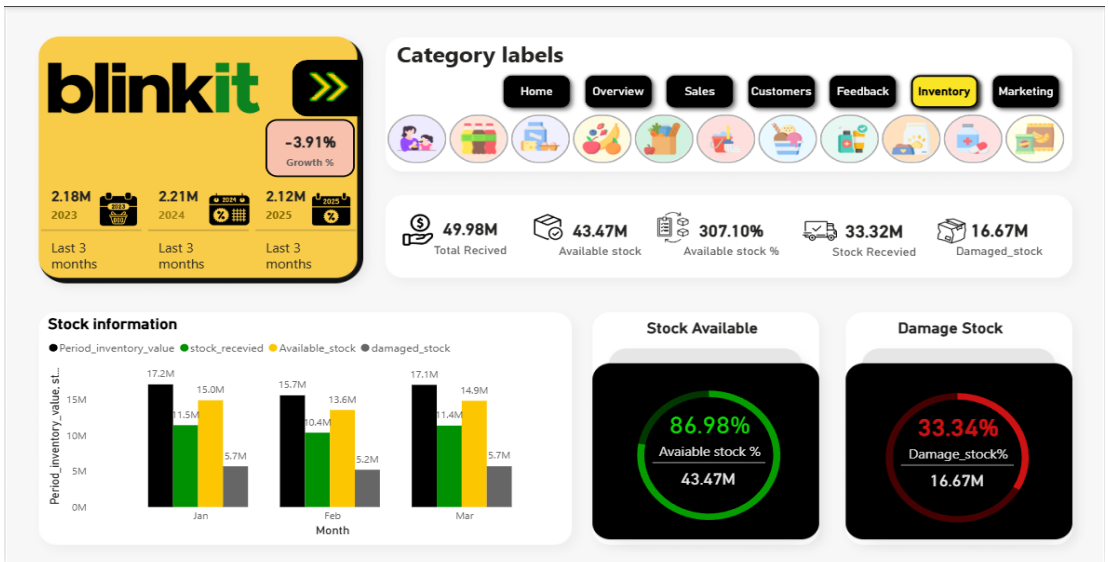
12.5 Inventory Page

Purpose:
Monitor stock health and operational efficiency.

Key Insights:

- Total stock received vs damaged stock
- Inventory availability percentage
- Time-based inventory tracking
- Identification of high-demand products with stock risk

Supports **inventory planning and stock optimization**.



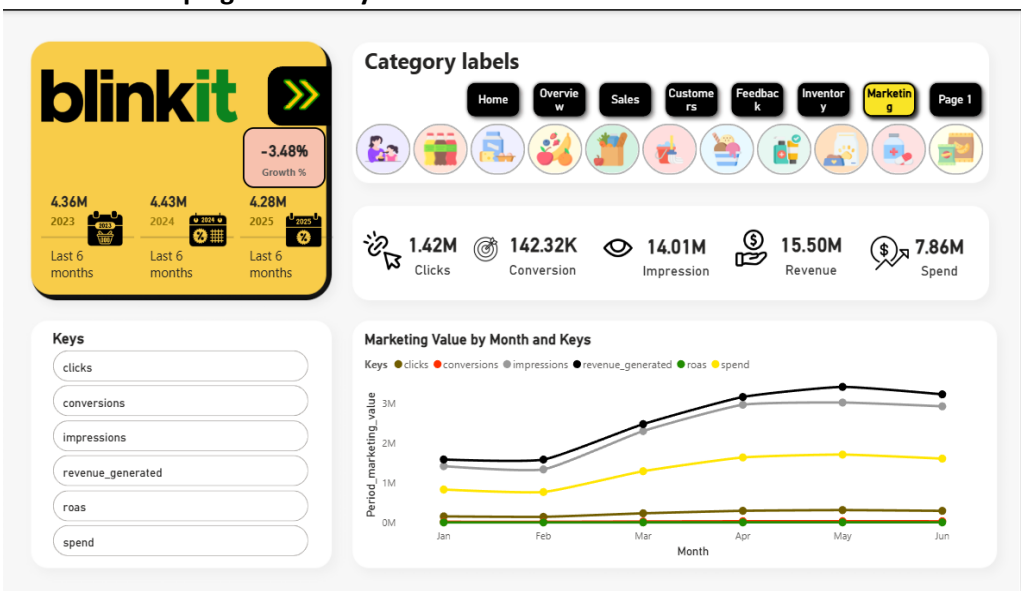
12.6 Marketing Page

Purpose:
Evaluate marketing campaign effectiveness.

Key Insights:

- Campaign-wise impressions, clicks, and conversions
- Revenue generated vs marketing spend
- ROAS analysis to identify high-performing campaigns
- Monthly marketing performance trends

Helps assess **ROI and campaign efficiency**.

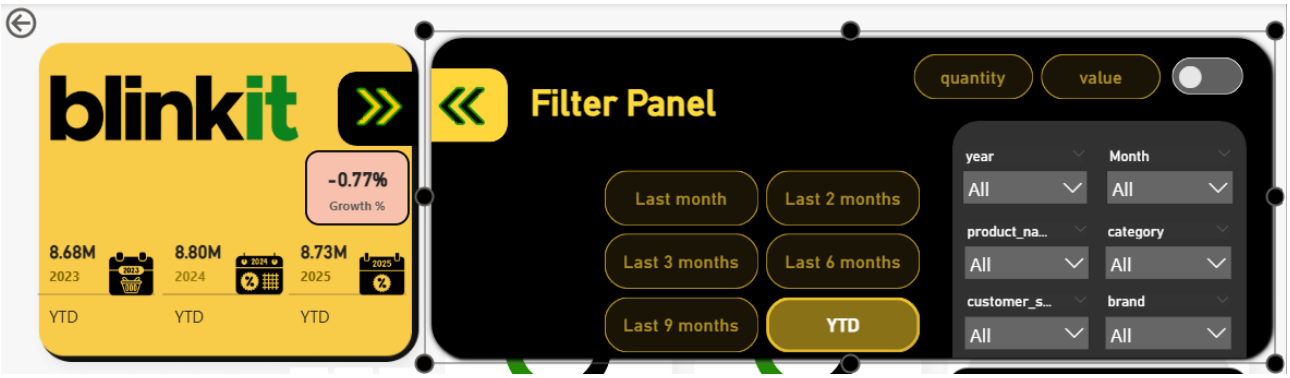


12.7 Filters & Interactivity

- Dynamic slicers for:
    - Date (Year, Month)

    - Product Category
    - Brand
    - Customer Segment
- Toggle between **Quantity vs Value**
- All visuals are interconnected for drill-down analysis

Ensures **flexibility and deep exploration**.



12.8 Business Value of the Dashboard

This Power BI dashboard enables:

- Faster decision-making
- Clear performance tracking
- Identification of growth and risk areas
- Alignment between sales, inventory, customer feedback, and marketing

12.9 Conclusion

The Power BI dashboard successfully transforms complex multi-source data into **actionable business insights** through intuitive visuals, filters, and KPIs.
It completes the analytics pipeline by turning analysis into **storytelling and decision support**.

13. Overall Project Conclusion

This project delivered an **end-to-end data analytics solution** for Blinkit by transforming raw operational data into **actionable business insights** using a structured analytical workflow.

The analysis began in **Jupyter Notebook**, where multiple datasets related to customers, orders, products, inventory, delivery, marketing, and feedback were cleaned, validated, and consolidated. Two key analytical tables were created:

- **Blinkit Master Table** for sales, customers, products, and feedback
- **Inventory Master Table** for stock and damage tracking

This ensured a clean, reliable, and scalable foundation for analysis.

Through **Exploratory Data Analysis (EDA)**, key patterns were identified:

- Customer ratings were predominantly positive, with **4-star ratings most frequent**
- Order values were concentrated within a specific range, indicating stable spending behavior
- Certain categories such as **Baby Care, Instant & Frozen Food, and Cold Drinks & Juices** consistently outperformed others
- Sentiment analysis revealed a **strongly positive customer perception**, with relatively fewer negative feedback instances

The **SQL analysis phase** translated these observations into structured business questions. SQL queries were used to:

- Identify top revenue-generating customers and product categories
- Analyze payment preferences and customer segments
- Detect high-risk scenarios such as low-stock high-demand products
- Evaluate delivery delays and their impact on customer ratings
- Identify churn-risk customers and profit leakage areas

These queries enabled precise, repeatable, and auditable insights directly from the database layer.

Finally, the **Power BI dashboard** transformed analytical findings into an **interactive decision-support system**. The dashboard provided:

- High-level business overviews
- Sales and revenue trend analysis
- Customer growth and churn insights
- Feedback and sentiment tracking
- Inventory health monitoring
- Marketing campaign performance and ROI analysis

Dynamic filters, slicers, and visual storytelling ensured that stakeholders could explore insights intuitively and in real time.

Overall, the project successfully aligned **business objectives with data-driven analysis**, delivering meaningful insights across sales, customer experience, operations, inventory, and marketing.

14. Future Scope of the Project

The current analysis provides a strong foundation; however, the project can be further enhanced in the following ways:

- **Real-time data integration** to enable live dashboards for operations and inventory
- **Predictive analytics** for:
    - Demand forecasting
    - Customer churn prediction
    - Inventory replenishment planning

- **Advanced customer segmentation** using clustering techniques
- **Delivery performance optimization** using route and time analysis
- **Automated alert systems** for:
    - Low stock levels
    - Negative feedback spikes
    - Delivery delays
- **Integration with machine learning models** for personalized recommendations
- Expansion to include **supplier performance and pricing optimization**

15. Final Outcome

This project demonstrates the complete analytics lifecycle:

- Data preparation (Python)
- Business querying (SQL)
- Insight visualization (Power BI)

It reflects strong analytical thinking, business understanding, and technical execution.
The solution is **scalable, interpretable, and aligned with real-world business needs**, making it suitable for professional, academic, and portfolio purposes.